

A2 Level Computer Science — Semester 1 Assessment Pack | Victoria Park Academy

Victoria Park Academy

A2 Level Computer Science — Semester 1 Assessment Pack

Academic Year 2024 – 2025

A2 Level Computer Science — Semester 1 Assessment Pack

Total Assessment Marks: 180 (Quizzes 85 + Mock Exam 75 + Practical Investigation 20)

Assessment Weighting: Quizzes 35% | Mock Exam 40% | Practical Investigation 25%

A2 计算机科学 — 第一学期评估包 / A2 Computer Science — Semester 1 Assessment Pack

本评估包涵盖数据库、布尔逻辑、排序与验证、网络、硬件和人工智能。This assessment pack covers databases, Boolean logic, sorting and validation, networks, hardware, and artificial intelligence.

Assessment Calendar — Semester 1

Week	Assessment	Topic	Marks	Duration	Format
Week 3	Quiz 1: Boolean & Floating-Point	Boolean algebra, K-maps, IEEE 754	20	25 min	Closed-book
Week 6	Quiz 2: Networks & Routing	TCP/IP, routing algorithms, DNS	20	25 min	Closed-book
Week 9	Formative Test: Weeks 1 – 8	Data Representation & Communication	40	90 min	Closed-book
Week 11	Quiz 3: Hardware & Adders	Logic circuits, half-adders, full-adders	20	25 min	Closed-book
Week 14	Quiz 4: System Software	Memory management, scheduling	20	25 min	Closed-book

Week	Assessment	Topic	Marks	Duration	Format
Week 17	Quiz 5: AI & Ethics	Expert systems, neural networks, ethics	20	25 min	Closed-book
Week 18	Mock Exam: Paper 1 Style	All Semester 1 topics	75	1h 45m	Closed-book

EAL Support Checklist — Assessment Preparation

Bilingual glossary (80+ terms) provided 1 week before each assessment

Extra reading time: +5 minutes for 25-minute quizzes, +15 minutes for mock exam

Visual diagrams included for all hardware and network questions

Command words defined: define, explain, compare, evaluate, design, trace

Sentence starters provided for extended-answer questions

Oral explanation option available for practical demonstrations

Quiz 1: Boolean Algebra & Floating-Point (20 Marks, 25 Minutes)

Topic Coverage: Boolean laws, De Morgan's, Karnaugh maps, IEEE 754 floating-point representation, normalisation, error calculation.

Question 1 — Boolean Simplification (6 marks)

1.1 Simplify the following Boolean expression using algebraic laws. Show each step clearly.

$$F = A \cdot B + A \cdot B' + A' \cdot B$$

(3 marks)

1.2 Use De Morgan's laws to simplify the expression below. Show your working.

$$F = (A + B)' \cdot (A \cdot B)'$$

(3 marks)

Question 2 — Karnaugh Map (6 marks)

2.1 A Boolean function has the following truth table:

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0

A	B	C	F
1	1	0	1
1	1	1	1

Construct a 3-variable Karnaugh map and derive the minimal Sum-of-Products expression. (4 marks)

2.2 Draw the logic circuit for your simplified expression using AND, OR, and NOT gates. (2 marks)

Question 3 — Floating-Point (5 marks)

3.1 Convert the denary number +6.5 to IEEE 754 single-precision floating-point format. Show all steps: binary conversion, normalisation, exponent with bias, and final 32-bit pattern. (4 marks)

3.2 A floating-point representation uses 8 bits for the mantissa and 4 bits for the exponent (both two's complement). State the smallest positive normalised number that can be represented. (1 mark)

Question 4 — Error Calculation (3 marks)

A number is stored as 0.1010×2^4 (4-bit mantissa). The exact value is 5.125.

4.1 Calculate the absolute error. (1 mark)

4.2 Calculate the relative error as a percentage. (2 marks)

Mark Scheme — Quiz 1: Boolean & Floating-Point

Q1 Boolean Simplification (6 marks)

1.1 $F = A \cdot B + A \cdot B' + A' \cdot B = A \cdot (B + B') + A' \cdot B = A \cdot 1 + A' \cdot B = A + A' \cdot B = A + B$ (1 mark per correct step, max 3)

1.2 $F = (A + B)' \cdot (A \cdot B)' = A' \cdot B' \cdot (A' + B') = A' \cdot B' \cdot A' + A' \cdot B' \cdot B' = A' \cdot B' + A' \cdot B' = A' \cdot B'$ (1 mark per correct step, max 3)

Q2 Karnaugh Map (6 marks)

2.1 Correct K-map populated (2 marks); correct grouping (1 mark); minimal SOP: $F = B + C$ (1 mark)

2.2 OR gate with B and C as inputs (2 marks)

Q3 Floating-Point (5 marks)

3.1 $6.5 = 110.1_{\text{2}} = 1.101 \times 2^3$ (1 mark); sign = 0, exponent = $2 + 127 = 129 = 10000001_{\text{2}}$ (1 mark); mantissa = 1010000000000000000000 (1 mark); final pattern assembled (1 mark)

3.2 0.10000000×2^4 (or equivalent) (1 mark)

Q4 Error (3 marks)

4.1 Stored value = $0.1010 \times 2^4 = 5.0$; Absolute error = $|5.125 - 5.0| = 0.125$ (1 mark)

4.2 Relative error = $0.125 / 5.125 = 0.02439... \approx 2.44\%$ (2 marks)

Quiz 2: Networks & Routing (20 Marks, 25 Minutes)

Topic Coverage: TCP/IP stack, routing algorithms (distance-vector, link-state, Dijkstra's), DNS, email protocols, HTML/CSS basics.

Question 1 — TCP/IP Protocol Stack (5 marks)

1.1 State the four layers of the TCP/IP model in order from highest to lowest. (2 marks)

1.2 For each layer, name one protocol that operates at that layer. (2 marks)

1.3 Explain the process of encapsulation as data moves from the application layer to the link layer. (1 mark)

Question 2 — Routing Algorithms (7 marks)

2.1 Explain the difference between distance-vector and link-state routing algorithms. (2 marks)

2.2 State one advantage of link-state routing over distance-vector routing. (1 mark)

2.3 Use Dijkstra's algorithm to find the shortest path from node A to all other nodes in the following network. Show your working in a table.

Edge	Weight
A - B	4
A - C	2
B - C	1
B - D	5
C - D	8
C - E	10
D - E	2

(4 marks)

Question 3 — DNS and Email (5 marks)

3.1 Explain the hierarchical structure of DNS, naming the levels from top to bottom. (2 marks)

3.2 Describe the difference between POP3 and IMAP for email retrieval. (2 marks)

3.3 State the purpose of an MX record in DNS. (1 mark)

Question 4 — HTML/CSS (3 marks)

4.1 Write the HTML code to create a heading "Welcome to VPA" and a paragraph describing the Computer Science department. (2 marks)

4.2 Write CSS to make all paragraph text blue and 16 pixels in size. (1 mark)

Mark Scheme — Quiz 2: Networks & Routing

Q1 TCP/IP (5 marks)

1.1 Application, Transport, Internet, Link (2 marks; 0.5 each)

1.2 One correct protocol per layer (2 marks; 0.5 each)

1.3 Each layer adds header information as data passes down (1 mark)

Q2 Routing (7 marks)

2.1 Distance-vector: routers share entire routing table with neighbours (1 mark); Link-state: routers share link information, each builds complete topology map (1 mark)

2.2 Any valid advantage: faster convergence, no count-to-infinity, more accurate (1 mark)

2.3 Correct Dijkstra's table (2 marks); correct shortest paths: A→B=3, A→C=2, A→D=7, A→E=9 (2 marks)

Q3 DNS & Email (5 marks)

3.1 Root, TLD, authoritative, subdomain (2 marks; 0.5 each)

3.2 POP3 downloads and typically deletes emails from server (1 mark); IMAP keeps emails on server and syncs across devices (1 mark)

3.3 Specifies mail server responsible for accepting email on behalf of a domain (1 mark)

Q4 HTML/CSS (3 marks)

4.1 Correct and tags with content (2 marks)

4.2 `p { color: blue; font-size: 16px; }` (1 mark)

Quiz 3: Sorting, Validation & Error Detection (20 Marks, 25 Minutes)

Topic Coverage: Sorting algorithms (merge sort, quick sort), validation techniques, parity checking, checksums, Hamming codes.

Question 1 — Sorting Algorithms (8 marks)

1.1 Describe how merge sort works, including the divide and conquer approach. (3 marks)

1.2 Trace merge sort on the array [8, 3, 6, 2, 7, 1, 5, 4]. Show the divide steps and the merge steps. (4 marks)

1.3 State the time complexity of merge sort in the best, average, and worst cases. (1 mark)

Question 2 — Validation (6 marks)

2.1 A program accepts a student's age as input. Describe three different validation checks that could be applied. (3 marks)

2.2 Explain the difference between validation and verification. Give an example of each. (3 marks)

Question 3 — Error Detection (6 marks)

3.1 A 7-bit Hamming code is used to protect a 4-bit data word. The data is 1011. Calculate the parity bits (even parity) and write the complete 7-bit code. Show your working. (4 marks)

3.2 Explain why Hamming codes can detect and correct single-bit errors, whereas simple parity can only detect errors. (2 marks)

Mark Scheme — Quiz 3: Sorting, Validation & Error Detection

Q1 Sorting (8 marks)

1.1 Divide array into halves recursively until single elements (1 mark); merge pairs in sorted order (1 mark); repeat merging until fully sorted (1 mark)

1.2 Correct divide tree (2 marks); correct merge steps showing sorted subarrays (2 marks)

1.3 $O(n \log n)$ for all cases (1 mark)

Q2 Validation (6 marks)

2.1 Any three valid checks: range (e.g., 0–120), type (integer), presence (not empty), format (numeric only), length (1–3 digits) (3 marks)

2.2 Validation ensures data is reasonable/allowed (1 mark); verification ensures data is correctly entered/transferred (1 mark); valid example of each (1 mark)

Q3 Error Detection (6 marks)

3.1 p1 covers positions 1,3,5,7; p2 covers 2,3,6,7; p4 covers 4,5,6,7 (1 mark); data at positions 3,5,6,7 = 1,0,1,1 (1 mark); p1=0, p2=1, p4=0 (1 mark); complete code: 0,1,1,0,0,1,1 (1 mark)

3.2 Hamming uses multiple parity bits covering overlapping groups (1 mark); can identify exact error position and flip bit; simple parity only detects odd number of errors (1 mark)

Quiz 4: Networks & Protocols (20 Marks, 25 Minutes)

Topic Coverage: Network topologies, media access control, Ethernet, Wi-Fi, network security (firewalls, encryption, certificates), client-server vs peer-to-peer.

Question 1 — Network Topologies (5 marks)

1.1 Draw and label a star topology and a bus topology. (2 marks)

1.2 Compare star and bus topologies in terms of reliability and cost. (2 marks)

1.3 State one advantage of a mesh topology over a star topology. (1 mark)

Question 2 — Network Security (7 marks)

2.1 Explain the purpose of a firewall in a network. (2 marks)

2.2 Distinguish between symmetric and asymmetric encryption. Give an example of when each would be used. (3 marks)

2.3 Explain how digital certificates are used to verify the identity of a website. (2 marks)

Question 3 — Client-Server vs P2P (5 marks)

3.1 Describe the client-server model of networking. (2 marks)

3.2 Describe the peer-to-peer (P2P) model of networking. (2 marks)

3.3 Give one advantage and one disadvantage of P2P compared to client-server. (1 mark)

Question 4 — Media Access Control (3 marks)

4.1 Explain how CSMA/CD works in Ethernet networks. (2 marks)

4.2 Explain why CSMA/CD is not needed in full-duplex switched Ethernet. (1 mark)

Mark Scheme — Quiz 4: Networks & Protocols

Q1 Topologies (5 marks)

1.1 Correct diagrams with labels (2 marks)

1.2 Star: more reliable (single point of failure is central node), higher cost (more cable) (1 mark);
Bus: less reliable (single cable failure), lower cost (1 mark)

1.3 Any valid advantage: multiple paths, no single point of failure, higher redundancy (1 mark)

Q2 Security (7 marks)

2.1 Monitors incoming/outgoing traffic (1 mark); blocks unauthorised access based on rules (1 mark)

2.2 Symmetric: same key for encryption and decryption, fast, used for bulk data (1 mark); Asymmetric: public/private key pair, slower, used for key exchange and digital signatures (1 mark); valid example of each (1 mark)

2.3 Certificate contains public key and identity signed by trusted CA (1 mark); browser verifies CA signature to confirm website identity (1 mark)

Q3 Models (5 marks)

3.1 Central server provides resources/services; clients request and receive (2 marks)

3.2 Each peer acts as both client and server; no central authority (2 marks)

3.3 Advantage: no single point of failure, scalable, lower infrastructure cost; Disadvantage: security concerns, inconsistent availability, harder to manage (1 mark for valid pair)

Q4 MAC (3 marks)

4.1 Listen before transmitting, detect collisions, back off and retry (2 marks)

4.2 Full-duplex uses separate channels for send/receive; no shared medium means no collisions (1 mark)

Mock Exam: Paper 1 Style (75 Marks, 1 Hour 45 Minutes)

Examination: A2 Level Computer Science — Paper 1 Mock

Time Allowed: 1 hour 45 minutes

Total Marks: 75

Instructions: Answer ALL questions. Write your answers in the spaces provided. Calculators are permitted. Show all working for calculation questions.

Section A: Short Answer (25 marks)

Question 1 — Boolean Algebra (4 marks)

1.1 Simplify the following expression using Boolean algebra laws. Show each step.

$$F = (A + B) \cdot (A + B')$$

(2 marks)

1.2 Use a Karnaugh map to simplify the function $F(A, B, C, D) = \sum (0, 2, 5, 7, 8, 10, 13, 15)$. Show your K-map and the simplified expression. (2 marks)

Question 2 — Floating-Point (5 marks)

2.1 A floating-point number is stored with an 8-bit two's complement mantissa and a 4-bit two's complement exponent. The stored pattern is:

Mantissa: 0.1101000

Exponent: 0011

Calculate the denary value. Show your working. (3 marks)

2.2 Explain why normalisation is important in floating-point representation. (2 marks)

Question 3 — Networking (5 marks)

3.1 Describe the function of the transport layer in the TCP/IP model. (2 marks)

3.2 Explain the difference between a router and a switch in a network. (2 marks)

3.3 State the purpose of the DNS protocol. (1 mark)

Question 4 — Hardware (5 marks)

4.1 Draw the truth table for a full-adder. (2 marks)

4.2 Explain how a ripple-carry adder is constructed from full-adders. (2 marks)

4.3 State one disadvantage of a ripple-carry adder compared to a carry-lookahead adder. (1 mark)

Question 5 — System Software (6 marks)

5.1 Explain the difference between paging and segmentation in memory management. (3 marks)

5.2 A system uses paging with a page size of 1024 bytes. A logical address is 3075. Calculate the page number and offset. (2 marks)

5.3 Explain what happens during a page fault. (1 mark)

Section B: Structured Problem Solving (30 marks)

Question 6 — Logic Circuit Design (8 marks)

A security system has three sensors: A (door), B (window), and C (motion). The alarm should sound when:

The door AND window are both triggered, OR

The motion sensor is triggered AND at least one of door or window is triggered

6.1 Write the Boolean expression for the alarm output. (2 marks)

6.2 Simplify the expression using Boolean algebra or a Karnaugh map. (3 marks)

6.3 Draw the logic circuit for the simplified expression using AND, OR, and NOT gates. (3 marks)

Question 7 — Dijkstra's Algorithm (8 marks)

The following weighted graph represents a network of routers:

Edge	Weight
A - B	2
A - C	6
B - C	3
B - D	1
C - D	4
C - E	5
D - E	3

7.1 Use Dijkstra's algorithm to find the shortest path from A to E. Show your working in a table. (5 marks)

7.2 State the shortest path and its total weight. (2 marks)

7.3 Explain why Dijkstra's algorithm is classified as a greedy algorithm. (1 mark)

Question 8 — Memory Management (7 marks)

A computer has 64KB of physical memory and uses paging with 4KB pages. The page table for a process is:

Page	Frame	Valid
0	5	1

Page	Frame	Valid
1	2	1
2	—	0
3	7	1

8.1 How many frames are available in physical memory? (1 mark)

8.2 Translate the following logical addresses to physical addresses, or state if a page fault occurs:

2048

8192

12288

(4 marks)

8.3 Explain how virtual memory allows a process to use more memory than is physically available. (2 marks)

Question 9 — CPU Scheduling (7 marks)

Four processes arrive at time 0 with the following burst times:

Process	Burst Time
P1	8
P2	4
P3	9
P4	5

9.1 Draw the Gantt chart and calculate the average waiting time for FCFS scheduling. (3 marks)

9.2 Draw the Gantt chart and calculate the average waiting time for SJF scheduling. (3 marks)

9.3 State which algorithm gives the lower average waiting time and explain why. (1 mark)

Section C: Extended Response (20 marks)

Question 10 — Artificial Intelligence and Ethics (10 marks)

A local council is considering implementing an AI system to screen job applications. The system uses machine learning trained on historical hiring data to predict which candidates are most likely to succeed.

10.1 Explain how a neural network could be trained to perform this classification task. (4 marks)

10.2 Identify and explain three ethical concerns with using this AI system for hiring decisions. (6 marks)

Question 11 — System Design and Evaluation (10 marks)

A university wants to design a new student registration system that must handle 10,000 concurrent users during peak registration periods. The system must be secure, reliable, and scalable.

11.1 Describe two hardware features that would improve the reliability of the system. (2 marks)

11.2 Explain how the system could use client-server architecture to support scalability. (3 marks)

11.3 Evaluate the use of cloud computing versus on-premises infrastructure for this system. Your answer should consider cost, security, and scalability. (5 marks)

Mark Scheme — Mock Exam Paper 1

Section A (25 marks)

1.1 $F = (A + B) \cdot (A + B') = A + B \cdot B' = A + 0 = A$ (1 mark per step, max 2)

1.2 Correct 4-variable K-map (1 mark); correct grouping (0.5 mark); simplified expression: $F = B'D' + BD$ (0.5 mark)

2.1 Mantissa = 0.1101000 = +0.8125; Exponent = 0011 = +3; Value = $0.8125 \times 2^3 = 6.5$ (1 mark per step, max 3)

2.2 Maximises precision by ensuring most significant bit is used (1 mark); ensures unique representation (1 mark)

3.1 Provides end-to-end communication (1 mark); TCP reliable/ordered, UDP fast/unreliable (1 mark)

3.2 Router: forwards between networks using IP addresses (1 mark); Switch: forwards within LAN using MAC addresses (1 mark)

3.3 Translates domain names to IP addresses (1 mark)

4.1 Full-adder truth table: 8 rows, sum and carry correct for all (2 marks)

4.2 Full-adders chained, carry-out of one to carry-in of next (2 marks)

4.3 Slower due to carry propagation delay (1 mark)

5.1 Paging: fixed-size pages, no external fragmentation (1 mark); Segmentation: variable-sized logical segments, matches program structure (1 mark); comparison of fragmentation or table structure (1 mark)

5.2 Page number = 3, offset = 3 (2 marks)

5.3 Required page not in memory, OS fetches from disk (1 mark)

Section B (30 marks)

6.1 Alarm = $(A \cdot B) + (C \cdot (A+B))$ (2 marks)

6.2 Simplified: Alarm = $A \cdot B + A \cdot C + B \cdot C$ or equivalent (2 marks); working shown (1 mark)

6.3 Correct circuit diagram (3 marks)

7.1 Correct Dijkstra's table (3 marks); correct shortest path calculation (2 marks)

7.2 $A \rightarrow B \rightarrow D \rightarrow E$, weight = 6 (2 marks)

7.3 Makes locally optimal choice at each step (1 mark)

8.1 16 frames (1 mark)

8.2 2048 $\rightarrow$ page 2, frame —, page fault (1 mark); 8192 $\rightarrow$ page 2, frame —, page fault (1 mark); 12288 $\rightarrow$ page 3, frame 7, offset 0, physical = 28672 (2 marks)

8.3 Pages stored on disk, loaded on demand (1 mark); page table maps to disk locations (1 mark)

9.1 FCFS Gantt: P1(0–8), P2(8–12), P3(12–21), P4(21–26); waiting times: 0, 8, 12, 21; average = 10.25 (3 marks)

9.2 SJF Gantt: P2(0–4), P4(4–9), P1(9–17), P3(17–26); waiting times: 9, 0, 17, 4; average = 7.5 (3 marks)

9.3 SJF gives lower average waiting time; shorter jobs don't wait behind long jobs (1 mark)

Section C (20 marks)

10.1 Input features from applications (1 mark); labelled training data (historical hires) (1 mark); forward propagation through layers (1 mark); backpropagation to adjust weights (1 mark)

10.2 Any three valid concerns with explanation: bias in training data (2 marks), lack of transparency (2 marks), privacy of applicant data (2 marks), accountability for decisions (2 marks), automation of human judgment (2 marks) — max 6 marks

11.1 Any two: RAID for redundancy, UPS for power failure, ECC memory, dual power supplies (2 marks)

11.2 Multiple servers handle requests (1 mark); load balancer distributes traffic (1 mark); database replication for read scaling (1 mark)

11.3 Cost: cloud = OpEx, on-prem = CapEx (1 mark); Security: on-prem = more control, cloud = shared responsibility (1 mark); Scalability: cloud = elastic, on-prem = limited by hardware (1 mark); balanced evaluation with conclusion (2 marks)

Grade Boundaries

Quiz Grade Boundaries (Raw Marks)

Grade	Quiz 1 (20)	Quiz 2 (20)	Quiz 3 (20)	Quiz 4 (20)
A	16 – 20	16 – 20	16 – 20	16 – 20
B	14 – 15	14 – 15	14 – 15	14 – 15
C	12 – 13	12 – 13	12 – 13	12 – 13
D	10 – 11	10 – 11	10 – 11	10 – 11
E	8 – 9	8 – 9	8 – 9	8 – 9
U	0 – 7	0 – 7	0 – 7	0 – 7

Mock Exam Grade Boundaries (75 marks)

Grade	Marks	UMS Equivalent
A	60 – 75	80 – 100%
B	52 – 59	70 – 79%
C	45 – 51	60 – 69%
D	37 – 44	50 – 59%
E	30 – 36	40 – 49%
U	0 – 29	0 – 39%

Semester 1 Overall Grade Boundaries (180 marks)

Grade	Marks	Percentage
A	144 – 180	80 – 100%
B	126 – 143	70 – 79%

Grade	Marks	Percentage
C	108 – 125	60 – 69%
D	90 – 107	50 – 59%
E	72 – 89	40 – 49%
U	0 – 71	0 – 39%

Practical Investigation (20 Marks)

Deadline: Week 16 of Semester 1

Task: Conduct a practical investigation into one of the topics covered in Semester 1, producing a structured report with evidence of practical work.

Investigation Options

Option	Title	Practical Component
A	Boolean Logic Circuit Design	Design and simulate a combinational circuit in Logisim
B	Network Protocol Analysis	Capture and analyse packets using Wireshark
C	Memory Management Simulation	Implement paging address translation in Python
D	Neural Network Experiment	Train a simple perceptron on a dataset using Python

Report Structure and Marks

Component	Marks	Description
Introduction & Aims	2	Clear statement of investigation purpose and research question
Background Research	3	Relevant theory explained with citations
Methodology	3	Detailed description of practical approach and tools used
Practical Evidence	5	Screenshots, code, diagrams, or logs showing practical work
Results & Analysis	4	Data presented and interpreted with clear reasoning
Conclusion & Evaluation	3	Findings summarised, limitations acknowledged, improvements suggested

Bronze: Complete investigation with basic practical work; report follows structure with some evidence; limited analysis.

Silver: Thorough practical work with clear evidence; good understanding of theory; structured analysis with some evaluation.

Gold: Sophisticated practical work with original elements; excellent theoretical understanding; critical analysis; creative evaluation with realistic improvements.

EAL Practical Investigation Support

Investigation brief provided in bilingual format

Report template with section headings and sentence starters

Visual evidence requirements clearly specified

One-to-one methodology check-in before practical work begins

Draft feedback provided at Week 14

Assessment Policy and Procedures

Special Considerations

EAL students receive +25% extra time for all timed assessments

Bilingual glossaries (80+ terms) permitted in all closed-book assessments

Students with learning support plans may use assistive technology as specified

All assessment materials available in large print upon request

Rest breaks available for students with medical needs

Academic Integrity

All submitted work must be the student's own

Plagiarism in the practical investigation will result in a zero mark and disciplinary action

Collaboration is permitted for peer review only; all practical work and writing must be individual

External sources must be cited using Harvard referencing style

Use of AI tools (e.g., ChatGPT) must be declared and appropriately referenced

Feedback and Review

Marked quizzes returned within one week with written feedback and target grades

Mock exam results discussed in individual target-setting meetings within two weeks

Practical investigation draft feedback provided at Week 14 (formative checkpoint)

Parents receive assessment reports after each quiz cycle

Student-teacher conferencing available on request for all assessments

中英双语术语 / Bilingual Assessment Terms:

Boolean algebra = 布尔代数 | Karnaugh map = 卡诺图 | Floating-point = 浮点数 | Normalisation = 规范化 |
TCP/IP = 传输控制协议/网际协议 | Routing = 路由 | DNS = 域名系统 | Encryption = 加密 | Firewall = 防火墙
| Logic gate = 逻辑门 | Half-adder = 半加器 | Full-adder = 全加器 | Flip-flop = 触发器 | Microprocessor =
微处理器 | Paging = 分页 | Segmentation = 分段 | Virtual memory = 虚拟内存 | Scheduling = 调度 |

Interrupt = 中断 | Neural network = 神经网络 | Machine learning = 机器学习 | Big data = 大数据 | Ethics = 伦理 | Algorithmic bias = 算法偏见

Victoria Park Academy | A-Level Computer Science Department

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